Majorana Modes in the Machine

Week 6: From One Measurement to a Phase Diagram

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Week 6 Objective

Goal

Turn the Week 5 proof-of-principle into a diagnostic: sweep μ and ask whether a gate-measurable non-local correlator reconstructs the topological phase boundary.

- Week 5 measured one sweet-spot example at $t = \Delta$, $\mu = 0$.
- Week 6 varies μ across trivial, critical, and topological regimes.
- The target observable remains parity-preserving and non-local.
- The emphasis now shifts from solving the model to diagnosing phases operationally.

The Observable We Keep

$$O_{\mathrm{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$$

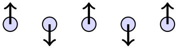
- Single-edge operators flip fermion parity, so their expectation vanishes in a parity eigenstate.
- Unlike Ising-type order parameters, topology cannot generally be detected locally.
- The Jordan–Wigner mapping converts an edge-to-edge Majorana correlator into a non-local Pauli string.
- The two-edge string preserves parity and probes coherent long-range edge correlations.
- For $L = 4$:

$$O_{\mathrm{edge}} = X_0 Z_1 Z_2 X_3.$$

Symmetry Breaking vs Topology

AFM transverse-field Ising chain

$$H = J \sum_j Z_j Z_{j+1} - h \sum_j X_j, \quad J > 0$$

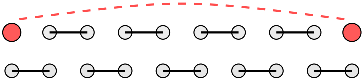


$$m_{\mathrm{stag}} = \frac{1}{L} \sum_j (-1)^j Z_j \neq 0$$

Two locally distinguishable Neel states; local domain-wall excitations.

Kitaev chain at $t = \Delta$

$$H = t \sum_j Y_j Y_{j+1} - \frac{\mu}{2} \sum_j Z_j$$



$$P = \prod_j Z_j, \quad O_{\mathrm{edge}} = X_0 Z \cdots Z X_{L-1}$$

Low-energy states differ by parity; Majorana edge degeneracy is non-local.

Same Ising-like algebra, different physics: local Neel order versus parity sectors and edge-to-edge coherence.

Measurement Protocol

Circuit-level measurement

- Current validation: exact even-parity ground state from diagonalization.
- Hardware target: parity-constrained VQE state preparation.
- Apply Hadamards on qubits carrying X factors.
- Measure all qubits and multiply outcomes to estimate $\langle O_{\mathrm{edge}} \rangle$.

Shot estimator ($N = 4096$)

Each shot returns an eigenvalue $m_s = \pm 1$.

$$\langle O \rangle \approx \frac{1}{N} \sum_{s=1}^{N} m_s, \quad \sigma \approx \sqrt{\frac{1 - \langle O \rangle^2}{N}}$$

For any ± 1 estimator with $N = 4096$, $\sigma \leq 1/\sqrt{N} \approx 0.016$. Error bars are omitted from the sweep plot for clarity.

Expected Physical Behavior

Topological

- Edge modes separated
- Finite string order
- Coherent boundaries

$$|\mu| < 2t$$

Critical

- Bulk gap closes
- Correlation length diverges
- Edge delocalization

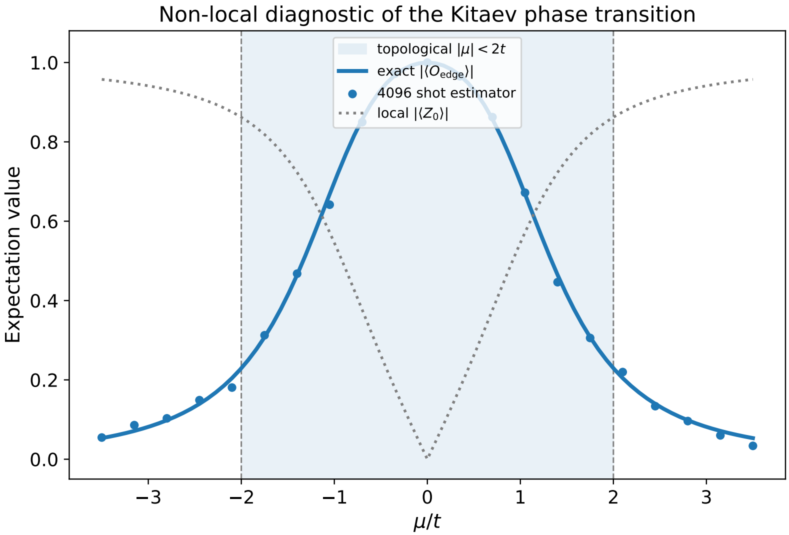
$$|\mu| = 2t$$

Trivial

- No protected localized zero modes
- Parity gap becomes finite
- Edge string becomes small

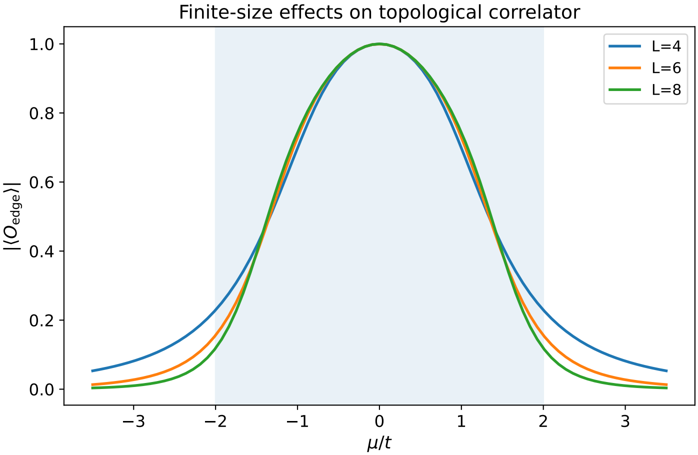
$$|\mu| > 2t$$

Numerical Phase Sweep



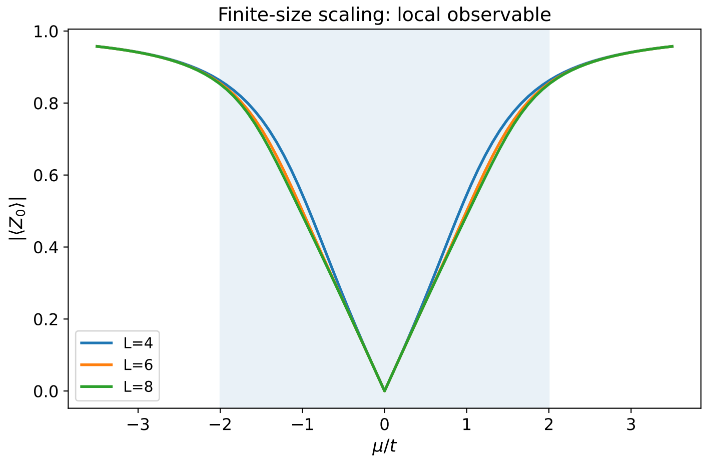
- $L = 4,\ t = \Delta = 1$; sweep μ
- Edge string: measurable diagnostic
- Parity gap: ED cross-check

Finite-Size Scaling: Non-Local Observable



- Small L: smooth crossover
- Larger L: sharper boundary
- $L = 4, 6, 8$ remain far from the thermodynamic limit
- The sharpening *trend*, not convergence, is the result

Finite-Size Scaling: Local Observable



- Local observable shows weak dependence on system size
- No sharpening near the phase transition
- Confirms that topology is not locally detectable

Implementation Step

```
for mu in mu_values:
    H = kitaev_qubit_hamiltonian(L, t, mu, delta)
    psi = even_parity_ground_state(H)    # exact diagonalization, NOT VQE
    value = <psi| X0 Z1 ... Z(L-2) X(L-1) |psi>
    shot_value = sample_pm_one(value, shots=4096)
```

- **Note:** `even_parity_ground_state` uses exact diagonalization – not VQE. The pseudocode masks this gap; a real VQE loop is the next milestone.
- Exact diagonalization is valid only for small L; it serves as a validation benchmark, not a scalable method.
- The measured quantity is the same Pauli string a circuit would estimate from bitstrings.
- The same workflow can be executed on quantum hardware using repeated circuit measurements.
- The next code revision should plug this sweep into the VQE runner directly.

Main Conclusions

- The edge Pauli string is a measurable finite-size diagnostic of Kitaev topology.
- Exact diagonalization shows string enhancement alongside near-degenerate parity sectors.
- Synthetic shot sampling reproduces the ideal string-correlation trend.
- A real VQE sweep and hardware-noise study remain the next validation steps.

Conceptual Message

Local observables may change across the transition, but parity-sector degeneracy and non-local edge coherence identify the topological phase.

Next: Block 4

- Add a real VQE loop for every μ point instead of ED-prepared validation states.
- Attach depolarizing, readout, and relaxation noise models to the measurement circuits.
- Compare ideal, noisy, and mitigated string-order curves.
- Study robustness of non-local topology diagnostics under realistic imperfections.
- Investigate how much noise the phase boundary tolerates before the signal becomes unreliable.
- Extend sweep to a 2D (t, μ) phase diagram.